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Characterization of α-glucosidase inhibitory constituents of the fruiting body of lion's mane mushroom (*Hericium erinaceus*)

Seul Ki Lee [a,1], Se Hwan Ryu [a,1], Ayman Turk [a], Sang Won Yeon [a], Yang Hee Jo [a], Yoo Kyong Han [b], Bang Yeon Hwang [a], Ki Yong Lee [b,**], Mi Kyeong Lee [a,*]

[a] College of Pharmacy, Chungbuk National University, Cheongju, 28160, Republic of Korea
[b] College of Pharmacy, Korea University, Sejong, 30019, Republic of Korea

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ABSTRACT

Ethnopharmacological relevance: Hericium erinaceus, commonly called lion's mane mushroom, is an edible and medicinal mushroom that has been traditionally used for the treatment of metabolic disorders, gastrointestinal diseases and memory impairment. In this study, potential anti-hyperglycemic constituents were identified to support the traditional usage of *H. erinaceus*.

Materials and methods: The components of *H. erinaceus* were purified using various column chromatography techniques. The structure of the separated compounds was determined based on spectroscopic data analysis, i.e., 1D and 2D NMR analysis. The anti-hyperglycemic activity of the isolated compounds was evaluated by measuring the inhibitory effects on α-glucosidase activity. Molecular docking analysis was also conducted for elucidation of α-glucosidase inhibitory activity of isolated compounds.

Results: Ten compounds including four new compounds, erinacenols A-D (**1**–**4**), were isolated from the fruiting bodies of *H. erinaceus.* Investigation of the anti-hyperglycemic effect of isolated compounds demonstrated that erinacenol D (**4**), 4-[3′,7′–dimethyl-2′,6′-octadienyl]-2-formyl-3-hydroxy-5-methyoxybenzylalcohol (**6**), hericene A (**7**), hericene D (**8**) and hericenone D (**9**) strongly inhibited α-glucosidase activity with IC_{50} values of <20 μM. The structure activity relationship suggested the importance of long side chain for α-glucosidase inhibitory activity. Further analysis by molecular docking demonstrated the interaction of α-glucosidase and isolated compounds, which supported the inhibitory activity of α-glucosidase.

Conclusion: Our present study demonstrated the beneficial effect of *H. erinaceus* by characterization of α-glucosidase inhibitory compounds, including four new compounds. This approach can be valuable support for the traditional use of *H. erinaceus* for the treatment of diabetes and metabolic diseases, which needs to be clarified by further in-vivo study.

1. Introduction

Diabetes mellitus is one of the most prevalent metabolic disorders in the world (Kazmi et al., 2018; Xu et al., 2018). It is characterized by hyperglycemia, mainly due to the imbalance between supply and demand of insulin. Hyperglycemia causes various complications and eventually develops into a fatal disease (Pinhas-Hamiel and Zeitler, 2007; Boyle, 2007). For the treatment of diabetes, various strategies have been developed to lower blood glucose. α-Glucosidase is a digestive enzyme for the degradation of carbohydrate. It catalyzes the final process of carbohydrate digestion by digesting carbohydrates into single monosaccharides; therefore, the inhibition of α-glucosidase can retard the absorption of carbohydrates in the small intestine and further reduce postprandial glucose and insulin levels (Xiao et al., 2013; Joshi et al.,

Abbreviations: NMR, nuclear magnetic resonance; IR, infrared; UV, ultraviolet; HR, high resolution; ESI, electrospray ionization; MS, mass; HSQC, heteronuclear single quantum coherence; HMBC, heteronuclear multiple bond correlation; HPLC, high performance liquid chromatography; MPLC, medium pressure liquid chromatography.

* Corresponding author.
** Corresponding author.
E-mail addresses: kylee11@korea.ac.kr (K.Y. Lee), mklee@chungbuk.ac.kr (M.K. Lee).
[1] These authors contributed equally.

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2015; Rahman et al., 2019). Practically, α-glucosidase inhibitors are used for the treatment of diabetes and other carbohydrate-mediated diseases including obesity. Based on this, natural products with α-glucosidase inhibition will be beneficial for the treatment and prevention of diabetes and metabolic diseases by controlling blood glucose levels (Joshi et al., 2015; Ríos et al., 2015).

Many traditional medicines have been used to treat diabetes for a long time. They have a therapeutic effect on diabetic symptoms and complications due to the combination of various components. Mushrooms are widely consumed for medicinal purposes. They are rich in nutrients and bioactive constituents, and thus, various pharmacological effects of mushrooms have been reported. Various mushrooms are used for the control of diabetes (Li et al., 2004; Nyam et al., 2017; Huang et al., 2018).

H. erinaceus is an edible and medicinal mushroom that belongs to the family Hericiaceae. It is also known as lion's mane mushroom, monkey's head mushroom and bearded tooth mushroom due to its unique shape and white spines. *H. erinaceus* has been used for the treatment of metabolic disorders, gastrointestinal diseases and memory impairment (Khan et al., 2013; Wang et al., 2014; He et al., 2017). Recently, interest in *H. erinaceus* for diabetes has increased, and its hypoglycemic properties have been demonstrated in a diabetes model (Wang et al., 2005; Zhang et al., 2017; Liang et al., 2013). *H. erinaceus* also exerted anti-obesity effects and improved lipid metabolism (Hiraki et al., 2017; Hiwatachi et al., 2010). Our preliminary study also showed the inhibitory effects of *H. erinaceus* on α-glucosidase activity. This prompted us to characterize the active constituents of *H. erinaceus.* In order to find α-glucosidase inhibitory secondary metabolites from *H. erinaceus,* chromatographic separation was conducted, which led to the isolation of 10 compounds including four new ones. The inhibitory activity of isolated compounds was investigated in our assay system and further analyzed using molecular docking analysis.

2. Material and methods

2.1. General experimental

A Jasco DIP-1000 polarimeter was used for the measurement of optical rotations. A Jasco UV-550 and PerkinElmer model LE599 spectrometer were used for the measurement of UV and IR spectra, respectively. NMR spectra were recorded on a Bruker DRX 400 or 500 MHz spectrometer using methanol-d_4 as solvents. ESIMS data was obtained on VG Autospec Ultima Mass spectrometers. HPLC grade solvents (methanol, acetonitrile, and water) were purchased from Fisher Scientific Korea, Ltd. α-Glucosidase (from *Saccharomyces cerevisiae*), *p*-nitrophenyl α-$_D$-glucopyranoside, and acarbose were purchased from Sigma Aldrich Co.

2.2. Plant material

The fruiting bodies of *H. erinaceus* were provided from the C&G Agricultural Association (Sejong, Korea). After identification by the herbarium of the College of Pharmacy, Chungbuk National University, voucher specimens (CBNU2017-HEF) were deposited in a specimen room of the herbarium.

2.3. Extraction and isolation

The dried fruiting bodies of *H. erinaceus* (3.8 kg) were pulverized and extracted twice with EtOAc. The EtOAc extract (180 g) was suspended in a mixture of CH_2Cl_2–MeOH–H_2O (10:9:1) and partitioned with *n*-hexane to give *n*-hexane fraction.

The *n*-hexane fraction (HEH, 100.2 g) was subjected to MPLC over silica gel (*n*-hexane -EtOAc, 100:0 → 0:100) to afford 14 subfractions (HEH1 - HEH14). HEH7 was subjected to MPLC over reverse phase silica gel (MeOH–H_2O, 10:90 → 100:0) to give 10 subfractions (HEH7A-HEH7J). Compounds **7** (4.4 mg) and **8** (4.7 mg) were purified from HEH7B by semi-preparative HPLC eluting with MeCN. HEH8 was subjected to MPLC over reverse phase silica gel (MeOH–H_2O, 10:90 → 100:0) to give nine subfractions (HEH8A - HEH8I). Compounds **9** (17.4 mg) and **10** (12.9 mg) were purified from HEH8D by semi-preparative HPLC eluting with MeCN–H_2O (90:10). HEH10 was subjected to MPLC over reverse phase silica gel (MeOH–H_2O, 10:90 → 100:0) to give eight subfractions (HEH10A - HEH10H). Compounds **1** (2.2 mg), **3** (1.3 mg), and **4** (9.8 mg) were purified from HEH10D by semi-preparative HPLC eluting with MeCN–H_2O (40:60). HEH10E was subjected to semi-preparative HPLC eluting with MeCN–H2O (65:35) to give compounds **5** (22.6 mg) and **6** (10.2 mg). Compound **2** (1.9 mg) was purified from HEH13D by semi-preparative HPLC eluting with MeCN–H_2O (50:50).

Table 1
^{1}H NMR assignments for compounds **1**–**4** in CD_3OD.

no.	1[a]	2[a]	3[a]	4[a]
3	5.25 (2H, s)	5.26 (2H, s)	5.26 (2H, s)	5.26 (2H, s)
4	6.70 (1H, s)	6.70 (1H, s)	6.70 (1H, s)	6.70 (1H, s)
1′	3.35 (2H, d, 7.5)	3.40 (2H, overlap)	3.36 (2H, d, 7.5)	3.36 (2H, d, 7.5)
2′	5.19 (1H, t, 7.5)	5.21 (1H, t, 7.5)	5.21 (1H, t, 7.5)	5.21 (1H, t, 7.5)
4′	1.96 (2H, m)	2.26, 1.97 (each 1H, m)	2.64 (2H, d, 5.5)	2.66 (2H, d, 4.5)
5′	1.57 (2H, m)	1.32 (2H, m)	5.56 (1H, m)[b]	5.56 (1H, m)[b]
6′	3.92 (1H, m)	3.19 (1H, m)	5.55 (1H, m)[b]	5.55 (1H, m)[b]
8′	4.75, 4.84 (each 1H, brs)	1.13 (3H, s)	1.24 (3H, s)	1.26 (3H, s)
9′	1.77 (3H, s)	1.78 (3H, s)	1.75 (3H, s)	1.75 (3H, s)
10′	1.66 (3H, s)	1.09 (3H, s)	1.24 (3H, s)	1.26 (3H, s)
OCH_3	3.90 (3H, s)	3.90 (3H, s)	3.91 (3H, s)	3.90 (3H, s)

[a] 500 MHz.
[b] Overlap.

Table 2
^{13}C NMR assignments for compounds **1**–**4** in CD_3OD.

no.	1[a]	2[a]	3[b]	4[a]
1	172.5	174.1	173.9	172.5
3	70.0	71.4	71.1	70.0
3a	147.5	148.9	148.9	147.6
4	96.1	97.5	97.5	96.1
5	164.7	166.2	166.1	164.7
6	116.6	118.2	118.0	116.5
7	154.1	155.7	155.5	154.0
7a	104.0	105.5	105.4	104.0
1′	21.1	22.6	22.6	21.2
2′	122.0	123.3	123.9	122.6
3′	134.4	136.3	135.2	133.6
4′	35.4	37.9	43.5	42.3
5′	32.9	30.7	126.1	128.2
6′	74.7	79.0	140.5	135.4
7′	147.4	73.8	71.4	81.0
8′	110.1	25.7	30.0	23.5
9′	16.1	16.3	16.3	14.9
10′	14.8	24.9	30.0	23.5
OCH_3	55.3	56.8	56.7	55.3

[a] 100 MHz.
[b] 125 MHz.

2.3.1. Erinacenol A (1)

Light brown syrup; UV (MeOH) λ_{max}: 218, 260 nm; IR ν_{max}: 3364, 1644 cm^{-1}; ^{1}H and ^{13}C-NMR spectral data, see Tables 1 and 2; positive ESIMS *m/z* 355 $[M+Na]^+$; HRESI-TOF-MS *m/z* 355.1514 $[M+Na]^+$ (calcd for $C_{19}H_{24}NaO_5^+$, 355.1516).

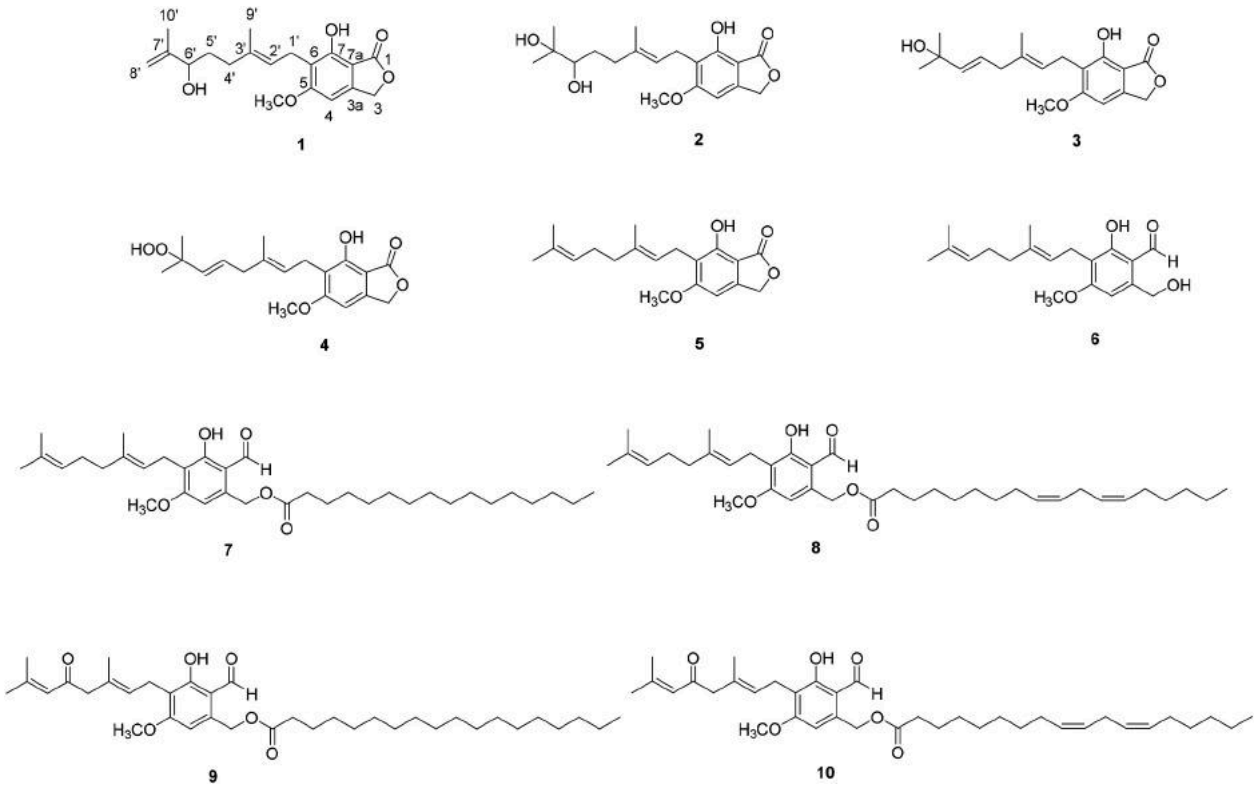

Fig. 1. Chemical structures of compounds **1**–**10** from *H. erinaceus*.

*2.3.2. Erinacenol B (**2**)*

Yellow syrup; UV (MeOH) λ_{max}: 218, 261 nm; IR ν_{max}: 3367, 1641 cm^{-1}; ^{1}H and ^{13}C-NMR spectral data, see Tables 1 and 2 positive ESIMS *m/z* 373 $[M+Na]^+$; HRESI-TOF-MS *m/z* 373.1621 $[M+Na]^+$ (calcd for $C_{19}H_{26}NaO_6^+$, 373.1622).

*2.3.3. Erinacenol C (**3**)*

Light brown syrup; UV (MeOH) λ_{max}: 220, 260 nm; IR ν_{max}: 3324, 1733 cm^{-1}; ^{1}H and ^{13}C-NMR spectral data, see Tables 1 and 2 positive ESIMS *m/z* 355 $[M+Na]^+$; HRESI-TOF-MS *m/z* 355.1515 $[M+Na]^+$ (calcd for $C_{19}H_{24}NaO_5^+$, 355.1516).

*2.3.4. Erinacenol D (**4**)*

Light brown syrup; UV (MeOH) λ_{max}: 220, 260 nm; IR ν_{max}: 3340, 1670 cm^{-1}; ^{1}H and ^{13}C-NMR spectral data, see Tables 1 and 2 positive ESIMS *m/z* 371 $[M+Na]^+$; HRESI-TOF-MS *m/z* 371.1465 $[M+Na]^+$ (calcd for $C_{19}H_{24}NaO_6^+$, 371.1465).

2.4. α-Glucosidase inhibitory activity

Inhibitory effects of isolated compounds on α-glucosidase activity was evaluated using α-glucosidase (α-glucosidase from *Saccharomyces cerevisiae*). Briefly, the test sample was mixed with α-glucosidase (1.0 U/mL) and incubated at 37 °C for 15 min. Ten μL of *p*-nitrophenyl α-D-glucopyranoside solution was added and enzyme reaction was carried out at 37 °C for 20 min. The amount of *p*-nitrophenol cleaved by the enzyme was determined by measuring the absorbance at 405 nm in a 96-well microplate reader. The percentage of α-glucosidase inhibition of compounds was calculated as [(Absorbance without sample – absorbance with sample)/absorbance without sample] x 100. Acarbose was used as a positive control.

2.5. Molecular docking analysis

Molecular docking studies were conducted by using SYBYL-X software (version 2.1.1) with crystal structures of isomaltase from *Saccharomyces cerevisiae* (PDB-ID: 3A4A). The three-dimensional (3D) structures of ligands were prepared by Chem3D Pro (version 12.0) and saved in a.mol file. The regions of target protein were prepared with the Tripos force field and all the water molecules were removed. Conflicted side chain amide was fixed. Two factors involved in generation, threshold and bloat, were configured at 0.5 and 0, respectively. The docking was a guided "sanitize" preparation protocol in SYBYL-X 2.1.1. The results of the docking approach and the ligands were ranked using total scores. Discovery Studio 2019 Client was used to display the position of ligands from the protein-ligand complex.

3. Results and discussion

Chromatographic purification of *n*-hexane-soluble fractions of *H. erinaceus* yielded 10 compounds (Fig. 1). The structures were determined by spectroscopic data analysis and comparison with literature values.

3.1. Structure elucidation

Chromatographic separation of *n*-hexane fraction of *H. erinaceus* were resulted in isolation of four new compound (**1**–**4**) together with six known compounds (**5**–**10**).

Compound **1** was isolated and purified by light brown syrup and its molecular formula of $C_{19}H_{24}O_5$ was confirmed by HRESI-TOF-MS spectrum through *m/z* 355.1514 $[M + Na]^+$ (calcd for $C_{19}H_{24}NaO_5^+$ 355.1516). The IR spectrum showed typical absorption bands of hydroxyl and carbonyl groups at 3364 and 1644 cm^{-1}, respectively. The signals at [δ_H 6.70 (1H, s, H-4), 5.25 (2H, s, H-3); δ_C 172.5 (C-1), 70.0 (C-3), 147.5 (C-3a), 96.1 (C-4), 164.7 (C-5), 116.6 (C-6), 154.1 (C-7), 104.0 (C-7a)] in the ^{1}H and ^{13}C NMR spectrum together with the HMBC correlations of H-3/C-1, 3a, 7a and H-4/C-3, C-6 suggested the presence of an isobenzofuranone moiety (Wang et al., 2017). In addition, the presence of 10 additional carbon signals attributed to the two pairs of

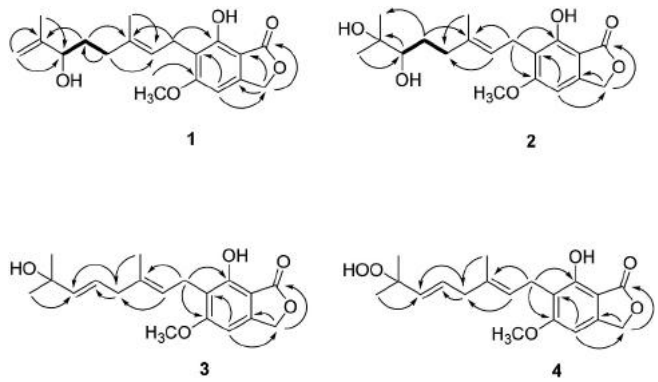

Fig. 2. Key HMBC (→) and COSY (▬) correlations of new compounds **1**–**4**.

isoprenyl derivatives and one methoxy signal suggested that compound **1** has a skeleton similar to hericenone J (**6**). Compared to hericenone J (6), the presence of a hydroxy group was also predicted by the signals at δ_H 5.55 (1H, m, H-6′) and δ_C 74.7 (C-6′) and signals attributed to the exomethylene group were also observed at δ_H 4.75, 4.84 (each 1H, brs, H-8′) and δ_C 110.1 (C-8′). The HMBC correlations of H-8’/C-6′, 7′, H-10’/C-6′ and H-5’/C-7′ determined the positions of exomethylene and hydroxyl groups, which was confirmed by the correlations between H-4’/H-5’/H-6′ in the ^{1}H–^{1}H COSY spectrum (Fig. 2). The position of the methoxy group was determined to be C-5 from the HMBC correlation. Based on the above analysis, the structure of compound **1** was determined as shown and named erinacenol A.

Compound **2** was isolated as a yellow syrup. The molecular formula of **2** was determined as $C_{19}H_{26}O_6$ from the HRESI-TOF-MS (*m/z* 373.1621 [M+Na]$^+$ (calcd for $C_{19}H_{26}NaO_6^+$ 373.1622). The ^{1}H and ^{13}C NMR spectra of **2** are quite similar to those of **1**, but signals for exomethylene were not observed in **2**. Instead, the presence of two methyl groups connected to hydroxymethine carbon were suggested from the signals at [δ_H 1.13, 1.09 (each 3H, s, CH_3-8′, 10′); δ_C 25.7, 24.9] together with δ_C 73.8 (C-7′). The HMBC correlations of CH_3-8’/C-6′ and H-6’/C-7′ determined the structure of **2** as shown and it was named erinacenol B.

Compound **3** was obtained as a light brown syrup. The molecular formula of $C_{19}H_{24}O_5$ was determined by the HRESI-TOF-MS ion at *m/z* 355.1515 [M+Na]$^+$ (calcd for $C_{19}H_{24}NaO_5^+$ 355.1516). The ^{1}H and ^{13}C NMR spectra of **3** are quite similar to those of **1** and **2**, which suggested that compound **3** has an isobenzofuranone structure. Comparing the ^{1}H and ^{13}C NMR spectra of compound **3** with those of **2**, it was presumed that a hydroxyl group at C-6′ of **2** was substituted with a double bond in **3** from the signals at [δ_H 5.55, 5.56 (each 1H, m); δ_C 126.1, 140.5], which were confirmed by HMBC correlations from H-4’/C-5′, 6′ and H-8’/C-6’.[23] Based on these data, the structure of compound **3** was defined as shown, and the compound was named erinacenol C.

The ^{1}H and ^{13}C NMR spectra of compound **4** were almost identical to those of compound **3** except for the downfield shifted chemical resonance at δ_C 81.0 (C-7′) in **4** compared to that of the oxygenated carbon at δ_C 71.4 (C-7′) in **3**. HRESI-TOF-MS analysis suggested the molecular formula of **4** as $C_{19}H_{24}O_6$, which has one more oxygen compared to **3**, confirmed the presence of a hydroperoxy group in the structure. Based on these data, the structure of compound **4** was determined as the hydroperoxy derivative of compound **3** and named erinacenol D.

The known compounds were identified as hericenone J (**5**), 4-[3′,7′–dimethyl-2′,6′-octadienyl]-2-formyl-3-hydroxy-5-methyoxybenzylalcohol (**6**), hericene A (**7**), hericene D (**8**), hericenone D (**9**), and hericenone E (**10**), by the analysis of their spectroscopic data and comparison with literature values (Ma et al., 2010; Li et al., 2017; Kawagishi et al., 1991; Miyazawa et al., 2012).

Table 3
Effect of compounds **1**–**10** on α-glucosidase activity.

Compounds	α-glucosidase inhibition		Docking analysis (Total score)
	inhibition at 100 μM.	IC_{50} (μM)	
1	58.6 ± 3.4	>20	7.561
2	17.0 ± 6.3	>20	9.035
3	42.7 ± 6.6	>20	6.518
4	82.4 ± 7.4	19.6	10.222
5	44.7 ± 8.6	>20	6.682
6	84.3 ± 6.5	7.5	7.840
7	91.4 ± 0.3	6.7	10.246
8	96.9 ± 1.6	3.9	10.525
9	89.2 ± 1.5	15.5	9.953
10	Not Tested	Not Tested	7.586
Acarbose[a]	78.6 ± 3.4	71.2	12.168

[a] Acarbose was used as the positive control.

Table 4
Molecular docking analysis of compounds **1**–**10** on α-glucosidase activity.

Compounds	Docking analysis (Total score)[a]	Key residues
1	7.561	Glu411(1.96), His351(2.83), Asp352(2.08, 2.19)
2	9.035	Glu411(1.99), Asp352(2.36)
3	6.518	Gln182(3.08), Gln353(2.49, 1.81), Asp215 (1.82)
4	10.222	Gln353(2.25), Arg442(1.83, 2.97), Asp307 (1.98), Asp69(2.13)
5	6.682	Asn350(2.87), Gln353(2.35), Glu277(1.78)
6	7.840	Gln182(2.26), Arg446(2.14), Asp215(2.20, 3.05), Asp69(1.92), Glu277(2.56, 2.38)
7	10.246	Arg442(2.37), Glu411(2.30), Tyr158(2.98, 3.09)
8	10.525	Gln353(1.89), Asp307(2.42, 2.49), Tyr158 (2.49), Glu411(2.51)
9	9.953	Asp307(2.19)
10	7.586	Gln353(2.22), Arg442(2.28, 2.68), Glu411 (2.53)
Acarbose[b]	12.168	Arg442(1.86, 2.88), His351(2.08, 2.22), Asp352(2.10, 2.18, 2.58), Glu277(2.17), Asp307(2.14), His280(1.91), Tyr158(2.28), Glu411(2.03, 1.90, 2.66, 2.66), Asp215(2.75, 2.51), Glu277(2.74), Asp69(2.87, 2.52, 2.73)

[a] Total scores were expressed in –log10(K_d)^2 units to represent binding affinities.
[b] Acarbose was used as the positive control.

3.2. α-Glucosidase inhibitory activity

α-Glucosidase is an intestinal enzyme involved in digestion and the absorption of carbohydrates by catalyzing the final step of carbohydrate digestion by converting carbohydrates to single monosaccharides. Therefore, the effect of isolated compounds on α-glucosidase inhibition was investigated. As shown in Table 3, a new compound, erinacenol D (**4**), together with 4-[3′,7′–dimethyl-2′,6′-octadienyl]-2-formyl-3-hydroxy-5-methyoxybenzylalcohol (**6**), hericene A (**7**), hericene D (**8**) and hericenone D (**9**) strongly inhibited α-glucosidase activity with IC_{50} values of <20 μM in our assay system. All compounds isolated from *H. erinaceus* share a structural similarity. Consistent with previous report of α-glucosidase inhibition of hericene A (Miyazawa et al., 2012), inference of a structural activity relationship suggested the importance of a long side chain for inhibition, as observed in compounds **7**–**9**. In addition, the polar side chains in compounds **4** and **6** also contributed to the inhibitory activity.

3.3. Molecular docking analysis

To support our results, molecular docking analysis was conducted. Consistent with our experimental results, compounds **4**, **7**–**9** had high

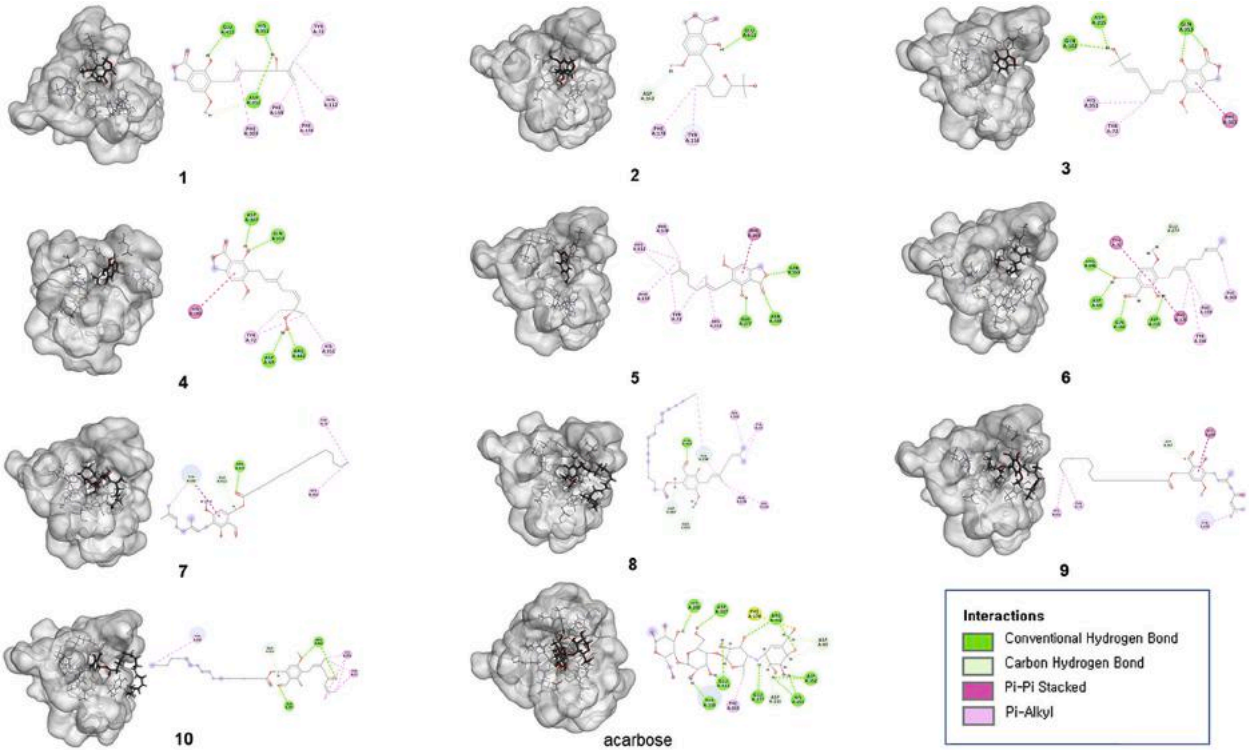

Fig. 3. Docking pictures of compounds **1**–**10**.

docking scores (Table 4). Molecular docking analysis also suggested the interaction modes, as illustrated in Fig. 3. Hydrogen bonds were formed between compound **8** with Gln353, Asp307, Tyr158 and Glu411, with lengths of 1.89, 2.42, 2.59 and 2.51Å, respectively (Fig. 3). Compound **4**, a new compound, formed hydrogen bonds with Gln353, Arg 442, Asp307 and Asp69, with lengths of 2.25, 1.83, 1.98 and 2.13 Å, respectively (Fig. 3). These results indicated that these compounds could be inserted into the active site of the enzyme and bind tightly to the catalytic amino acid residues, which could inhibit α-glucosidase activity.

4. Conclusion

H. erinaceus has been used for the regulation of metabolic diseases in traditional medicines. Therefore, we tried to find out the active compounds to support traditional usage. In this study, 10 aromatic compounds including four new compounds, erinacenols A-D (**1**–**4**), were isolated and their structures were elucidated based on spectroscopic analysis. Further biological assessment revealed the α-glucosidase inhibitory activity of the isolated compounds. In addition, binding of active compounds to the active site of α-glucosidase was suggested by molecular docking analysis. Taken together, these findings might explain the beneficial effects of *H. erinaceus* and its constituents against diabetes and other metabolic disease, which needs to be validated by further study using in-vivo model.

Author contributions

Mi Kyeong Lee planned and designed this research. Seul Ki Lee, Se Hwan Ryu, Ayman Turk, Sang Won Yeon, Yang Hee Jo and Yoo Kyong Han conducted the experiment. Mi Kyeong Lee, Ki Yong Lee and Bang Yeon Hwang analyzed and interpreted the data. Seul Ki Lee, Se Hwan Ryu and Mi Kyeong Lee wrote the manuscript. All authors read and approved the manuscript.

Declaration of competing interest

The authors declare no conflicts of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jep.2020.113197.

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